Exercises: Constrained Optimization and the Implicit Function Theorem

Mathematical Foundations for Machine Learning

If we are in f:Rp×RmRm then we can write every m input varin m output v

Implicit Function Theorem

Exercise 1.1 [Easy]
Consider the equation $F(x, y) = x^2 + y^2 - 1 = 0$ and the point $(x_0, y_0) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.

(a) Verify that $F(x_0, y_0) = 0$.

(b) Check the hypothesis of the Implicit Function Theorem: compute $\frac{\partial F}{\partial y}(x_0, y_0)$ and verify it is nonzero.

(c) Conclude that there exists a function $y = \varphi(x)$ defined near x_0 such that $F(x, \varphi(x)) = 0$. Find $\varphi'(x_0)$ by implicit differentiation.

From that we can reconstruct taylor because we get '(x)y(x0)

Exercise 1.2 [Medium]
Consider the system of equations $F : \mathbb{R}^3 \to \mathbb{R}$ defined by

$$F(x_1, x_2, x_3) = x_1^2 - x_1 x_3 + x_2^2 x_1 - x_2^2 = 0.$$

(a) Verify that $z = (1, 1, 1)$ is a solution.

(b) Can we locally solve for x_3 as a function of (x_1, x_2) near z? Justify using the Implicit Function Theorem.

(c) Compute $\frac{\partial x_3}{\partial x_1}$ and $\frac{\partial x_3}{\partial x_2}$ at the point $(1, 1)$ by implicit differentiation.

(d) At what points does the IFT fail to guarantee a local solution for x_3? Interpret geometrically.

Exercise 1.3 [Hard]
Let $F : \mathbb{R}^4 \to \mathbb{R}^2$ be defined by

$$F(x, y, u, v) = \begin{pmatrix} x^2 + y^2 - u^2 - v \\ xy - uv \end{pmatrix}.$$

Consider the point $z_0 = (1, 1, 1, 1)$.

(a) Verify that $F(z_0) = 0$.

(b) Can we locally solve for (u, v) as functions of (x, y) near z_0? Set up and verify the appropriate condition from the Implicit Function Theorem.

(c) Compute the Jacobian matrix $D_{(x,y)}(u, v)$ at $(1, 1)$ using the implicit differentiation formula:

$$D_y \varphi = -(D_y F)^{-1} \cdot D_x F,$$

where $\varphi(x, y) = (u(x, y),\ v(x, y))$.

Lagrange Multipliers — Setup and Computation

Exercise 2.1 [Easy]
Find the extrema of $f(x, y) = x + y$ subject to the constraint $x^2 + y^2 = 1$.

(a) Write down the Lagrangian $\mathcal{L}(x, y, \lambda)$.

(b) Compute the stationarity conditions $\nabla_{x,y}\mathcal{L} = 0$ and the constraint equation.

(c) Solve the system to find all critical points and classify them as maxima or minima.

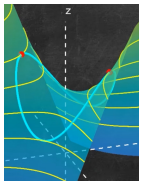
The gradient of the restriction g and the function to minimize f are parallel to each other and perpendicular to the tangent line.

Exercise 2.2 [Easy]
Find the point on the plane $2x + y + z = 5$ that is closest to the origin.

(a) Formulate this as a constrained optimization problem: state the objective function and the constraint.

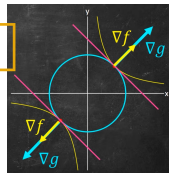
(b) Write the Lagrangian and derive the stationarity conditions.

(c) Solve for the closest point.



f(c)=g(c)

c is one of the two opt



Exercise 2.3 [Medium]
Consider the curve $y = 1.5 - x^2$ in $\mathbb{R}^2$.

(a) Formulate the problem of finding the point(s) on this curve closest to the origin as a constrained optimization problem using the squared Euclidean distance.

(b) Write down the Lagrangian. Derive the stationarity conditions.

(c) Solve the system. You should find that there are two cases depending on the value of λ. Find all constrained minimizers.

(d) State the necessary and sufficient conditions for a point to be a constrained minimizer in general.

We take the direction in which we fulfill the constraints: v,g(c)=0The hessiar

Exercise 2.4 [Medium]
Find the closest point to $(0, 0)$ in the set

$$M = \{x \in \mathbb{R}^2 : x_1 + x_2 \geq 4,\ 2x_1 + x_2 \geq 5\}.$$

(a) Formulate as an optimization problem with inequality constraints. Write the objective and constraints in the standard form $g_i(x) \leq 0$.

(b) Write the Lagrangian and state all four KKT conditions.

(c) Solve the KKT system. Use complementary slackness to enumerate cases.

Exercise 2.5 [Medium]
Consider the entropy maximization problem on the probability simplex:

$$\max_{x \in \mathbb{R}^n} \left(-\sum_{i=1}^{n} x_i \log x_i \right) \quad \text{s.t.} \quad \sum_{i=1}^{n} x_i = 1, \quad x_i \geq 0.$$

(a) Write the Lagrangian incorporating both the equality and inequality constraints.

(b) Derive the stationarity condition with respect to x_i.

(c) Using complementary slackness, argue that $x_i > 0$ for all i at the optimum. Then solve for x_i.

(d) What is the maximum entropy value?

Lagrange Multipliers — Proofs and Theory

Exercise 3.1 [Easy]
Let $g : \mathbb{R}^n \to \mathbb{R}$ be differentiable and $N = \{x : g(x) = 0\}$. Suppose $c \in N$ is a local minimum of a differentiable function f on N, and $\nabla g(c) \neq 0$.

(a) State the necessary condition (Lagrange condition) that must hold at c.

(b) Give a geometric interpretation: what does $\nabla f(c) = \lambda \nabla g(c)$ mean about the gradients at the constrained optimum?

(c) Why is the condition $\nabla g(c) \neq 0$ (constraint qualification) needed?

Exercise 3.2 [Medium]
Proving the Lagrange necessary condition in $\mathbb{R}^2$.

Let $f, g : \mathbb{R}^2 \to \mathbb{R}$ be C^1 functions, $N = \{(x, y) : g(x, y) = 0\}$, and suppose $c = (a, b) \in N$ is a local minimum of f on N with $\frac{\partial g}{\partial y}(c) \neq 0$.

(a) By the Implicit Function Theorem, there exists φ with $g(x, \varphi(x)) = 0$ near a. Define $h(x) = f(x, \varphi(x))$. Explain why $h'(a) = 0$.

(b) Compute $h'(a)$ using the chain rule and implicit differentiation (express $\varphi'(a)$ in terms of partial derivatives of g).

(c) Use parts (a) and (b) to show that $\nabla f(c) = \lambda \nabla g(c)$ for some $\lambda \in \mathbb{R}$.

Exercise 3.3 [Hard]
Sufficient condition for a constrained minimum.

Let $f, g : \mathbb{R}^n \to \mathbb{R}$ be C^2, $N = \{x : g(x) = 0\}$, and suppose at $c \in N$ there exists λ such that the Lagrangian $h = f - \lambda g$ satisfies $\nabla h(c) = 0$.

Define the **constrained tangent space** $T_c = \{v \in \mathbb{R}^n : \langle v, \nabla g(c) \rangle = 0\}$.

(a) State the sufficient condition for c to be a strict local constrained minimum in terms of the Hessian of h restricted to T_c.

(b) *Application.* Consider $f(x, y) = x^2 + 2y^2$ subject to $g(x, y) = x + y - 1 = 0$. Find the critical point using Lagrange multipliers. Then verify the sufficient condition to confirm it is a local minimum.

(c) *Application.* Consider maximizing $f(x, y, z) = xyz$ on the sphere $x^2 + y^2 + z^2 = 1$. Find the critical points in the first octant $(x, y, z > 0)$. Verify using the bordered Hessian or the restricted Hessian that this is indeed a local maximum.

Lagrange Multipliers with Inequality Constraints (KKT)

Exercise 4.1 [Easy]
State all four Karush–Kuhn–Tucker (KKT) conditions for the problem

$$\min_x f(x) \quad \text{s.t.} \quad g_i(x) \leq 0,\ i = 1, \ldots, m, \quad h_j(x) = 0,\ j = 1, \ldots, p.$$

Briefly explain the role of each condition.

Exercise 4.2 [Medium]
Minimize $f(x_1, x_2) = (x_1 - 3)^2 + (x_2 - 2)^2$ subject to

$$x_1 + x_2 \leq 4, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

(a) Rewrite the constraints in standard form $g_i(x) \leq 0$.

(b) Write the Lagrangian and the full set of KKT conditions.

(c) Solve the system. Enumerate cases using complementary slackness and find the optimal solution.

Exercise 4.3 [Hard]
Consider the KL-divergence minimization problem from the 2026 exam:

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^n x_i (\log x_i - \log p_i) \quad \text{s.t.} \quad \sum_{i=1}^n x_i = 1, \quad x_i \geq 0,$$

where $p_i > 0$ are given with $\sum_i p_i = 1$.

(a) Is the objective function convex? Justify.

(b) Is the objective function **strongly** convex? Prove or disprove.

(c) Assuming strong duality holds, write down the Lagrangian, the dual function (minimizing over x), and solve the dual problem.

(d) What is the optimal solution x^*? Interpret the result in terms of information theory.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
Spectral optimization via Lagrange multipliers.

Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix with eigenvalues $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$.

(a) Formulate the problem $\max_{\|x\|_2=1} x^T Ax$ as a constrained optimization problem and apply the method of Lagrange multipliers. Show that the critical points satisfy $Ax = \mu x$ for some multiplier μ, i.e., they are eigenvectors.

(b) Conclude that the maximum of the Rayleigh quotient $R_A(x) = \frac{x^T Ax}{x^T x}$ equals λ_n.

(c) Now consider the 2026 exam-style problem: $A = aa^T + bb^T$ with $a = u + v$, $b = \frac{3}{4}u - 3v$, where u, v are orthonormal. Find $\max_{\|x\|=1} \|Ax\|_2 - \langle x, Ax\rangle$.

(*Hint: First compute the eigenvalues of A, then express the objective in spectral coordinates.*)

Exercise 5.2 [Hard]
From constrained optimization to MLE.

Consider N data points $\{(x_n, y_n)\}_{n=1}^N$ with $x_n \in \mathbb{R}^D$, $y_n \in \mathbb{R}$, arising from $y = x^T\theta + \varepsilon$ where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ i.i.d.

(a) Write the conditional probability $P(Y|X;\theta)$ using the i.i.d. assumption.

(b) Show that the maximum likelihood estimate is $\hat{\theta} = (X^T X)^{-1} X^T Y$.

(c) Now suppose we add a regularization constraint $\|\theta\|_2^2 \le r^2$ (Ridge-style). Formulate the constrained MLE as a Lagrangian problem and use KKT conditions to show that the solution satisfies

$$\hat{\theta}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T Y$$

for some $\lambda \ge 0$.

(d) Interpret the role of λ: what happens when $\lambda = 0$ vs. $\lambda \to \infty$?

Exercise 5.3 [Hard]
Lagrange multipliers meet the Implicit Function Theorem.

Let $f, g : \mathbb{R}^3 \to \mathbb{R}$ be C^2 functions. Define the Lagrangian system $F : \mathbb{R}^4 \to \mathbb{R}^3$ by

$$F(x_1, x_2, x_3, \lambda) = \begin{pmatrix} \nabla f(x) - \lambda \nabla g(x) \\ g(x) \end{pmatrix} = \begin{pmatrix} \partial_1 f - \lambda \partial_1 g \\ \partial_2 f - \lambda \partial_2 g \\ \partial_3 f - \lambda \partial_3 g \\ g(x) \end{pmatrix} \in \mathbb{R}^4.$$

(Note: F maps $\mathbb{R}^4 \to \mathbb{R}^4$.)

Suppose (x^*, λ^*) is a solution of $F = 0$.

(a) Under what conditions on f and g can we apply the Implicit Function Theorem to conclude that the solution (x^*, λ^*) is locally unique (isolated)?

(b) Compute the Jacobian DF explicitly in terms of the Hessians of f and g and the gradient ∇g.

(c) *Specific case.* Let $f(x) = x_1^2 + x_2^2 + x_3^2$ and $g(x) = x_1 + x_2 + x_3 - 3 = 0$. Find the constrained minimum using Lagrange multipliers and verify that the Jacobian DF is invertible at the solution.